Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/07/2005
PI Name:	Chang, Hsueh-Chia Ph.D.		
Project Title:	Microcirculation Anomalies in Microgravity Blood Flow		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Hsueh-Chia.Chang.2@nd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	574 631 5697
Organization Name:	Notre Dame University		
PI Address 1:	165A Fitzpatrick Engr.		
PI Address 2:			
PI Web Page:			
City:	Notre Dame	**State:**	IN
Zip Code:	46556	**Congressional District:**	3
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/04/2002	**End Date:**	10/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Ansari, Rafat	**Contact Phone:**	216-433-5008
Contact Email:	Rafat.R.Ansari@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2701		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project is to study anomalies in micro-gravity blood flow in human circulation system and in blood diagnostic kits. We have solved an important blood suspension loading problem into small diagnostic kits and are now working on several chip-scale blood analytical tools. Included are blood coagulation detecion, blood glucose detection, leukemia, sickle cell and blood bacteria detection. The target is to produce simple kits that do not require elaborate equipment that is unavailable in space. To develop blood diagnostic kits that can be used in a confined spaceship. The diagnostic kits would be to detect pathogens and anomal physiological symptons duirng prolonged space flights. It can produce a robust and miniature bloood diagnostic kit for a variety of blood anomalies and diseases often associated with space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The same blood diagnostic kits could be used as point-of-care and portable diagnostic kits on earth.
Task Progress:	We have solved an important blood suspension loading problem: why blood suspension cannot invade into a microneedle less than 100 microns in diameter. The phenomenon is due to a blood cell migration mechanism driven by cell deformability. Blood cells migrate to the middle of the capillary and are endowed with a higher velocity than the meniscus velocity. As a result, they pack behind the wetting meniscus and the concentrated slug arrests further invasion of the blood suspension. We have proposed several microfluidic designs that can be used to prevent this loading failure. We have used the same migration mechanism to develop a kit that can measure the deformability of the blood cells. As leukemia cells, sickle cells and septic cells are less deformable than healthy blood cells, we are able to detect the former cells using the kits based on the above principles. We have also developed a microfluidic device that can separate blood cells from plasma. An impedance spectroscopy method was also developed to measure the coagulation rate of blood.
Bibliography Type:	Description: (Last Updated: 02/21/2014)
Articles in Peer-reviewed Journals	Wu J, Ben Y, Battigelli D, Chang HC. "Long-range AC electroosmotic trapping and detection of bioparticles." Industrial and Engineering Chemistry Research. 2005 Feb;44(8):2815-22. http://dx.doi.org/10.1021/ie049417u , Feb-2005
Papers from Meeting Proceedings	Zhou R, Chang BW, Chang HC. "Impedance Analysis of Blood Coagulation by Prothrombin Time Assay in a Miniature Device." ASME 3rd International Conference on Microchannels and Minichannels, Toronto, Ontario, Canada, June 13–15, 2005. ASME 3rd International Conference on Microchannels and Minichannels. Paper No. ICMM2005-75155, p. 737-741. http://dx.doi.org/10.1115/ICMM2005-75155 , Jun-2005